

March 3, 2026

Dr. Joel D. Stitzel

Editor-in-Chief

*Annals of Biomedical Engineering*

Dear Dr. Stitzel,

We submit our manuscript titled "*Video-based analysis of head-torso coupling during lateral impacts under passive and co-contracted conditions*" for consideration as an Original Article in the *Annals of Biomedical Engineering*. This manuscript has not been published previously and is not under consideration at another journal. All authors have approved the manuscript and agree with its submission.

Current anthropomorphic test devices and computational head-neck models lack empirical data on head-torso coupling during impacts. We addressed this gap by conducting a secondary video analysis of high-speed footage from Reynier et al. (2020), tracking simultaneous head and torso motion in 19 male participants during controlled lateral head impacts under passive and co-contracted cervical muscle conditions. We found that co-contraction reduced peak head displacement by 30% and head-torso relative excursion by 36% while leaving peak linear acceleration and angular kinematics unchanged. All 19 participants showed displacement reductions. These results demonstrate that cervical muscle activation constrains head displacement through increased mechanical coupling rather than through force attenuation.

These findings fill a specific gap in the surrogate design literature. No previous study has directly measured head-torso coupling through simultaneous tracking of both segments in living humans during lateral impacts. Our displacement, velocity, and coupling metrics define the response envelope that biofidelic surrogate necks must replicate across passive and actively braced conditions. The parent dataset (Reynier et al. 2020) was published in this journal, and several key studies on head impact biomechanics, wearable sensor validation, and surrogate neck design that motivate our work also appeared in ABME. We believe our manuscript aligns well with the journal's scope in injury biomechanics and biomedical device validation.

The original experimental protocol was approved by the University of Virginia Institutional Review Board (IRB-HSR #20486). All participants provided written informed consent for both the original data collection and future secondary analyses. Analysis code and processed summary data are publicly available on GitHub. The authors declare no competing interests.

We appreciate your consideration of this work and welcome any feedback from the editorial team.

Sincerely,

A handwritten signature in black ink, appearing to be 'Pawel Kudzia', with a stylized, cursive script.

**Pawel Kudzia**

On behalf of all authors

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